

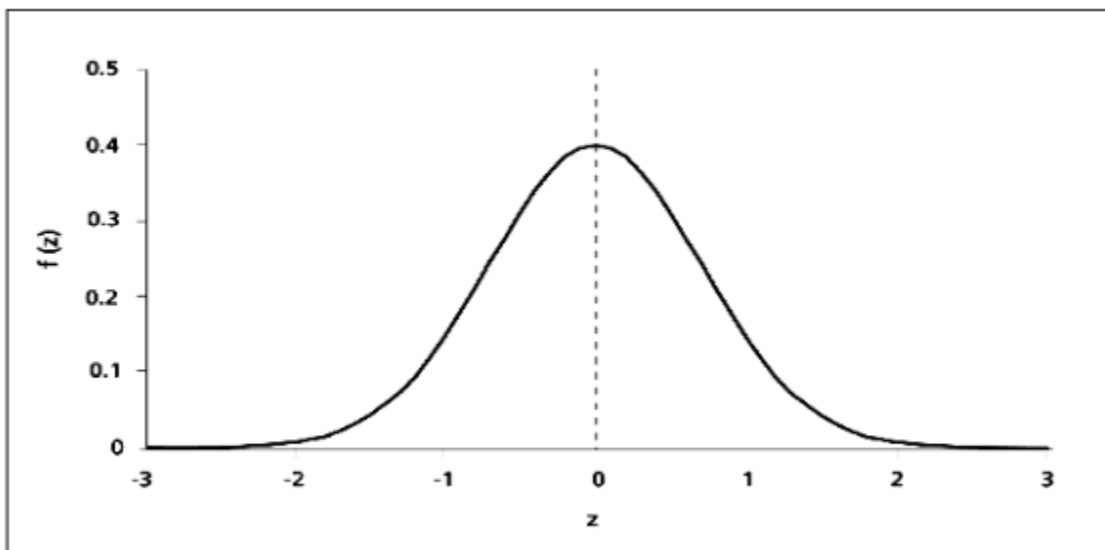
Distribution

Normal Distribution

The normal distribution, also known as the Gaussian distribution, is the most important probability distribution in statistics for independent, random variables. Most people recognize its familiar bell-shaped curve in statistical reports.

The normal distribution is a continuous probability distribution that is symmetrical around its mean, most of the observations cluster around the central peak, and the probabilities for values further away from the mean taper off equally in both directions.

A normal distribution is determined by two parameters the mean and the variance. A normal distribution with a mean of 0 and a standard deviation of 1 is called a standard normal distribution.



Distribution

Why is the normal distribution important?

The bell-shaped curve is a common feature of nature and psychology

The normal distribution is the most important probability distribution in statistics because many continuous data in nature and psychology displays this bell-shaped curve when compiled and graphed.

For example, if we randomly sampled 100 individuals we would expect to see a normal distribution frequency curve for many continuous variables, such as IQ, height, weight and blood pressure.

Converting the raw scores of a normal distribution to z-scores

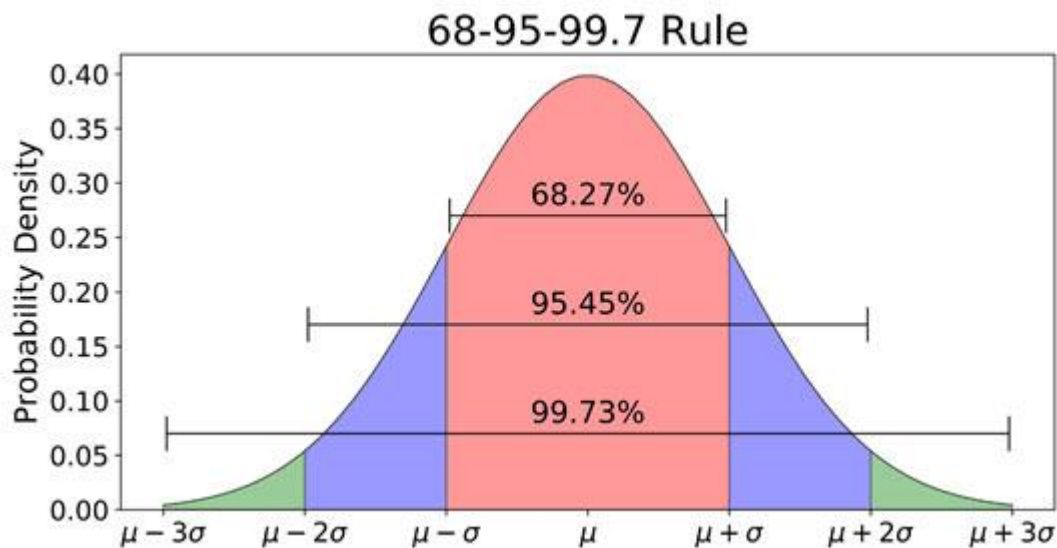
We can standardized the values (raw scores) of a normal distribution by converting them into z-scores.

This procedure allows researchers to determine the proportion of the values that fall within a specified number of standard deviations from the mean (i.e. calculate the empirical rule).

Distribution

Probability and the normal curve: What is the empirical rule formula?

The empirical rule in statistics allows researchers to determine the proportion of values that fall within certain distances from the mean. The empirical rule is often referred to as the three-sigma rule or the 68-95-99.7 rule.

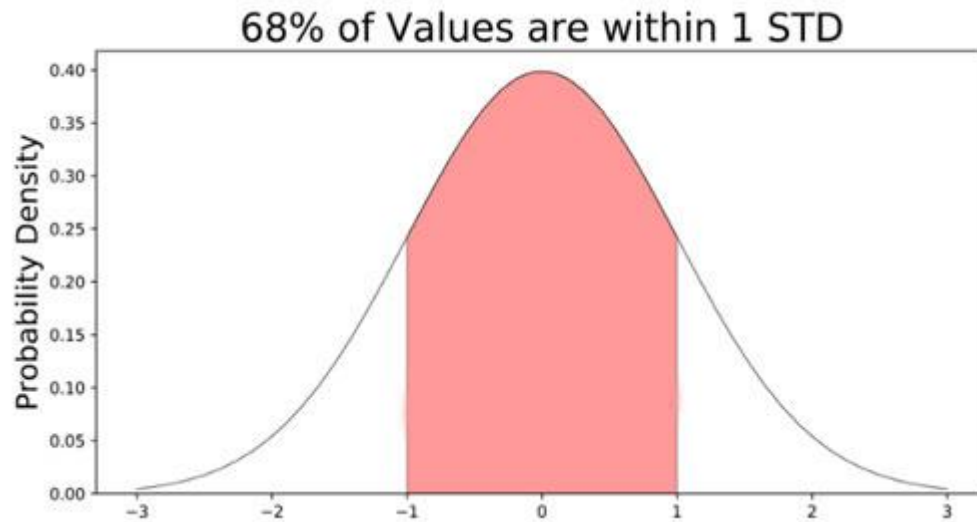


If the data values in a normal distribution are converted to [standard score](#) (z-score) in a standard normal distribution the empirical rule describes the percentage of the data that fall within specific numbers of standard deviations (σ) from the mean (μ) for bell-shaped curves.

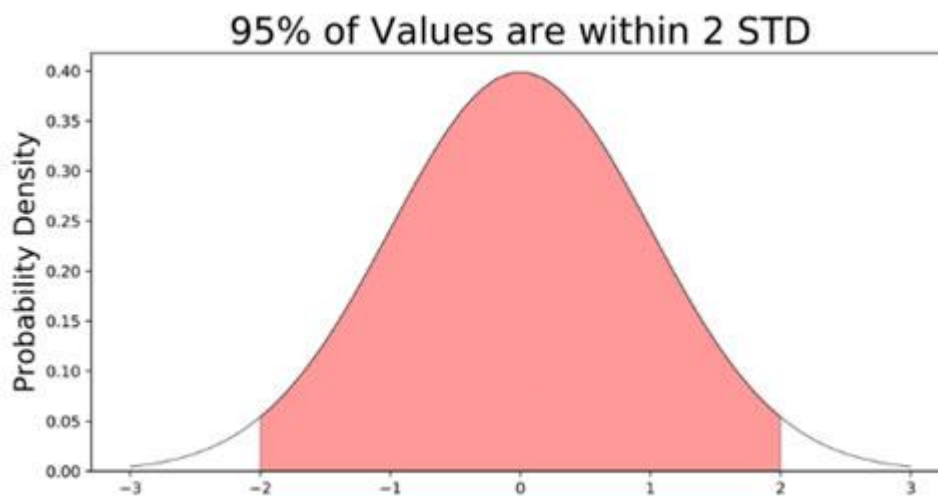
The empirical rule allows researchers to calculate the probability of randomly obtaining a score from a normal distribution.

68% of data falls within the first standard deviation from the mean. This means there is a 68% probability of randomly selecting a score between -1 and +1 standard deviations from the mean.

Distribution

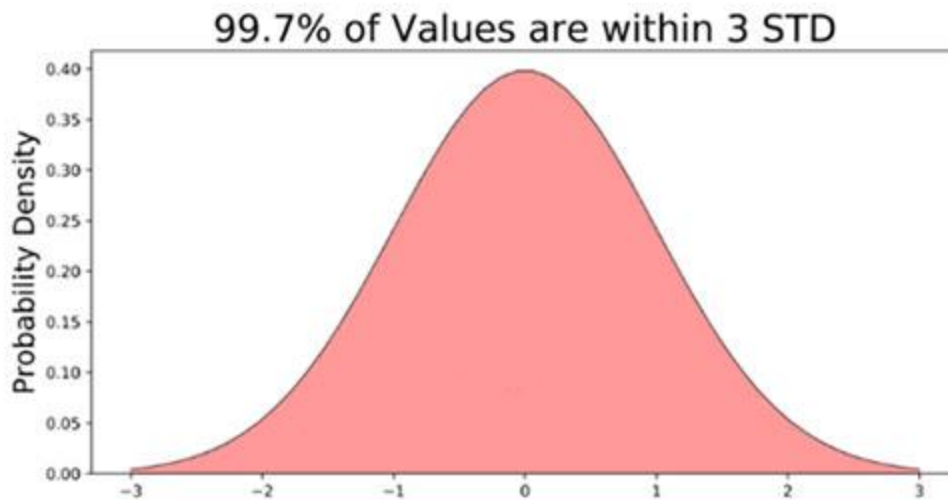


95% of the values fall within two standard deviations from the mean. This means there is a 95% probability of randomly selecting a score between -2 and +2 standard deviations from the mean.



Distribution

99.7% of data will fall within three standard deviations from the mean. This means there is a 99.7% probability of randomly selecting a score between -3 and +3 standard deviations from the mean.



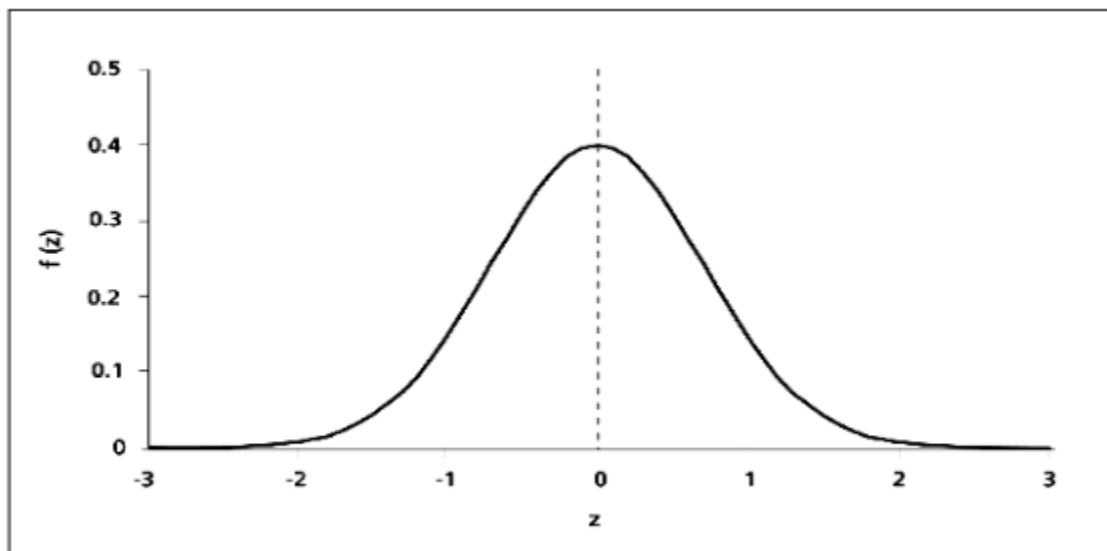
Distribution

Z-Score: Definition, Calculation and Interpretation

What does the z-score tell you?

z-score describes the position of a raw score in terms of its distance from the mean, when measured in standard deviation units. The z-score is positive if the value lies above the mean, and negative if it lies below the mean.

It is also known as a standard score, because it allows comparison of scores on different kinds of variables by standardizing the distribution. A standard normal distribution (SND) is a normally shaped distribution with a mean of 0 and a standard deviation (SD)



Why are z-scores important?

It is useful to standardized the values (raw scores) of a normal distribution by converting them into z-scores because:

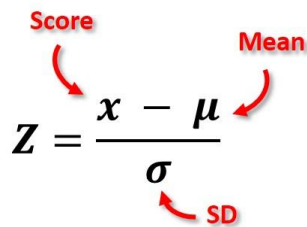
- (a) it allows researchers to calculate the probability of a score occurring within a standard normal distribution;
- (b) and enables us to compare two scores that are from different samples (which may have different means and standard deviations).

Distribution

How do you calculate the z-score?

The formula for calculating a z-score is $z = (x - \mu) / \sigma$, where x is the raw score, μ is the population mean, and σ is the population standard deviation.

As the formula shows, the z-score is simply the raw score minus the population mean, divided by the population standard deviation.



The diagram shows the formula $Z = \frac{x - \mu}{\sigma}$. Red arrows point from labels to the variables: 'Score' points to x , 'Mean' points to μ , and 'SD' (Standard Deviation) points to σ .

How do you interpret a z-score?

The value of the z-score tells you how many standard deviations you are away from the mean. If a z-score is equal to 0, it is on the mean.

A positive z-score indicates the raw score is higher than the mean average. For example, if a z-score is equal to +1, it is 1 standard deviation above the mean.

A negative z-score reveals the raw score is below the mean average. For example, if a z-score is equal to -2, it is 2 standard deviations below the mean.